

Fake News Detection: Semantic Classification Model

Assignment Title: Fake News Detection

Group Members:

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1. Introduction and Objectives

The goal of this assignment is to build a model that can classify news articles as real or fake using semantic information from the text. To do this, we used Word2Vec to capture the relationships between words, helping our machine learning models understand the content beyond just surface-level patterns. This is especially useful in tasks like fake news detection, where the meaning of words and context plays a big role in making accurate predictions.

From a practical point of view, the project addresses the growing issue of misinformation online. With the massive amount of news shared every day, it's become harder to tell reliable information from false or misleading stories. Our aim is to develop a system that can automatically flag fake news, helping reduce the spread of misinformation and support informed public opinion.

Here's the step-by-step process we followed to build the model:

1. Data Preparation
2. Text Preprocessing
3. Train/Validation Split
4. EDA on Training Data
5. EDA on Validation Data
6. Feature Extraction using Word2Vec
7. Model Training and Evaluation

2. Assumptions

While working on this project, we made the following assumptions:

- The labels in the dataset (True.csv and Fake.csv) are accurate and correctly represent whether each article is real or fake.
- All analysis and model training were done using only the provided data, without adding anything from outside sources.
- The insights and conclusions in this report are based entirely on patterns found

in this specific dataset.

- Model performance has been evaluated using only the training and validation sets. We haven't tested the models on any real-world or external data.

3. Data Preparation

The initial step involved loading True.csv (21,417 true news articles) and Fake.csv (23,523 fake news articles), each containing 'title', 'text', and 'date' columns.

- **Labelling and Merging:** To prepare for classification, we added a new column called "news_label" to each dataset — assigning 1 for true news and 0 for fake. After that, we combined both datasets into one "DataFrame" called "combined_news_df".
- **Handling Missing Values:** We checked for null values and found some missing entries in the title, text, and date columns. Since both title and text are important for our model, we dropped any rows where either of these was missing.
- **Combining Features:** To simplify the input, we merged the title and text columns into a new column named "news_text". This gave us a single content field per article. After creating this, we removed the original title, text, and date columns, keeping only "news_label" and "news_text" for the next steps.

4. Text Preprocessing

We applied several text preprocessing steps to the news_text column to get it ready for feature extraction and modeling:

1. **Lowercasing:** All the text was converted to lowercase so that words like "The" and "the" are treated the same.
2. **Removing Square Brackets:** We removed anything inside square brackets (like [Reuters]), since these usually contain metadata that doesn't add value to the content.
3. **Stripping Punctuation:** All punctuation marks were taken out to reduce noise and keep the focus on the words.
4. **Removing Words with Numbers:** Words that had numbers in them (like dates or numerical references) were filtered out, as they typically don't help in this kind of semantic analysis.
5. **Cleaning Extra Spaces:** Finally, we removed any extra spaces and trimmed the text to ensure it was clean and uniform.

6. **POS Tagging and Lemmatization:** After cleaning the text, we tokenized each article and tagged every word with its part of speech.

- Then, we applied lemmatization to reduce words to their base forms — for example, "running" became "run" and "better" became "good".
- To focus on the most meaningful content, we kept only the words that were tagged as nouns (both singular and plural forms) and filtered out common stopwords like "the," "is," and "and." This helped us zero in on the key concepts and entities in each article, which are often carried by nouns in the text.

Character Length Analysis (Validation Data):

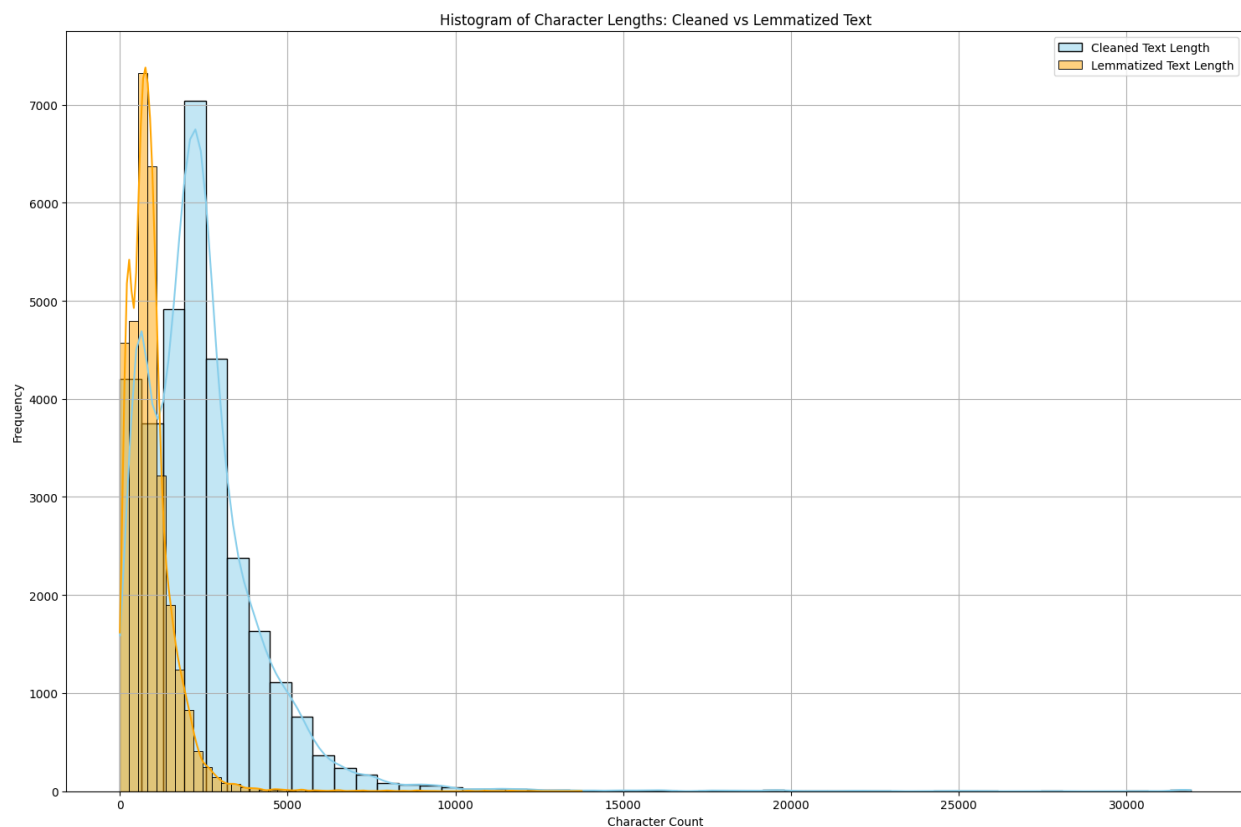


Figure 1: Histogram to visualize character lengths

Observation:

- This histogram shows the character length distribution for "Cleaned Text" and "Lemmatized Text" in the validation data.
- Both distributions are right-skewed.
- "Lemmatized Text" is significantly shorter, peaking around 500 characters, indicating effective length reduction by lemmatization.
- "Cleaned Text" has a broader, longer distribution, peaking between 1500-2500

characters.

- There is overlap, but lemmatization clearly condenses texts to shorter lengths.
- In summary, the histogram demonstrates that lemmatization substantially reduces text character length, shifting the distribution to the left.

5. Exploratory Data Analysis (EDA)

EDA was performed on both training and validation datasets to understand the characteristics of true and fake news.

5.1 EDA on Training Data

Top Words by Frequency (Word Clouds)

For True News:-



Figure 2: Top 40 Words in True News (Training Set) Word Cloud

Observation:

- The word cloud clearly highlights the most frequent words in true news articles within the training dataset after lemmatization.
- Key observations include:
 - Dominant terms like "state," "government," and "year" suggest a focus on official affairs, political structures, and temporal aspects.
 - Words such as "time," "day," "week," and "month" further reinforce the temporal nature of news reporting.

- For Fake News: -**



- This word cloud showcases the top 40 most frequent words found in fake news articles within the training dataset, after text processing.
- Key observations include:
 - The most striking feature is the overwhelming prominence of "trump," which is significantly larger than any other word, suggesting a strong focus on this

individual in fake news narratives.

- Other notable names like "clinton" and "obama" also appear, though much smaller, indicating that fake news often revolves around specific political figures.
- Words like "people," "time," "year," and "country" are also quite frequent, indicating a general news-like structure, but the context is likely skewed or exaggerated.
- Terms such as "government," "state," "house," "president," and "american" suggest a continued engagement with political and national themes.
- Interestingly, words like "fact," "report," "thing," "case," and "question" appear, which might be used in fake news to lend an air of legitimacy or to frame speculative content.
- Terms like "video," "family," and "campaign" hint at the diverse topics and methods used in fake news dissemination.
- In summary, the word cloud for fake news distinctively highlights a heavy emphasis on "trump" and other political figures, alongside general news vocabulary, but likely used in a misleading or sensationalized context, differentiating its thematic focus from true news.

N-gram Analysis (Bar Charts)

Top 10 Unigrams (True News - Training Set):

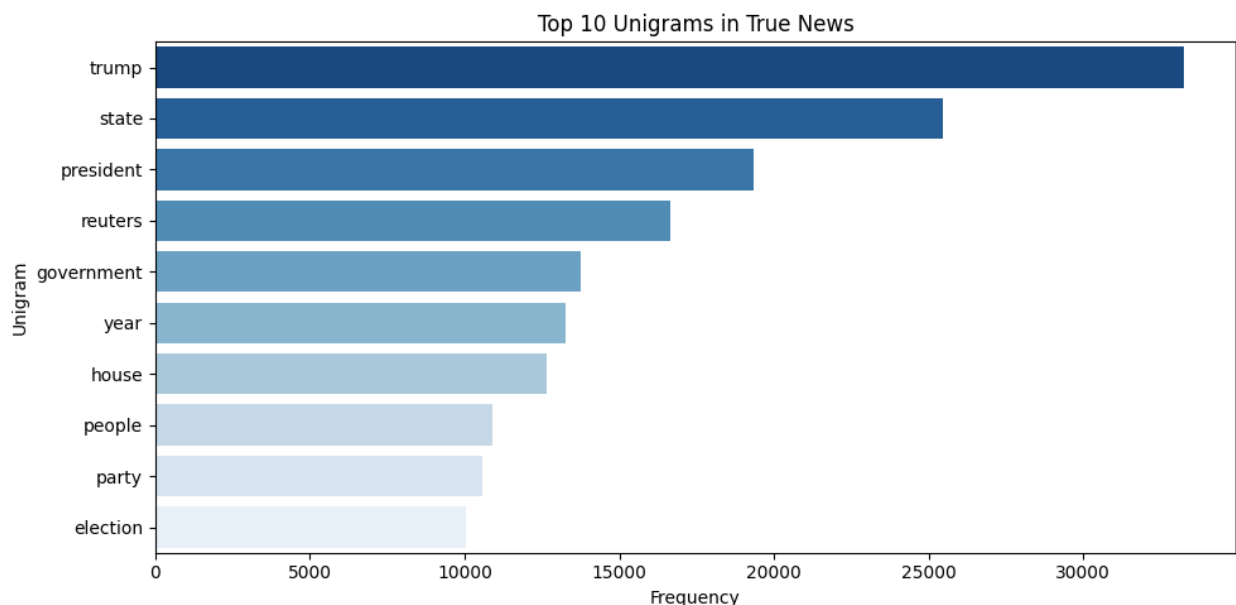


Figure 4: Top 10 Unigrams in True News (Training Set) Bar Chart

Observation:

- This horizontal bar chart presents the top 10 most frequent unigrams in true

news articles within the training set.

- Key observations include:
 - "trump" is by far the most frequent unigram, indicating a significant focus on this individual in true news reporting.
 - "state," "president," and "government" are also very high in frequency, reinforcing the emphasis on political and governmental topics.
 - "reuters" appearing as a top unigram suggests that Reuters news agency is a prominent source or is frequently mentioned within the true news articles.
 - Other frequent terms like "year," "house," "people," "party," and "election" further highlight the common themes of true news, which revolve around temporal aspects, political bodies, and general societal engagement.
- In summary, the unigram analysis of true news clearly points to a strong concentration on political figures and institutions, with "trump" being a central figure, and "reuters" identified as a significant entity within the dataset.

Top 10 Bigrams (True News - Training Set):

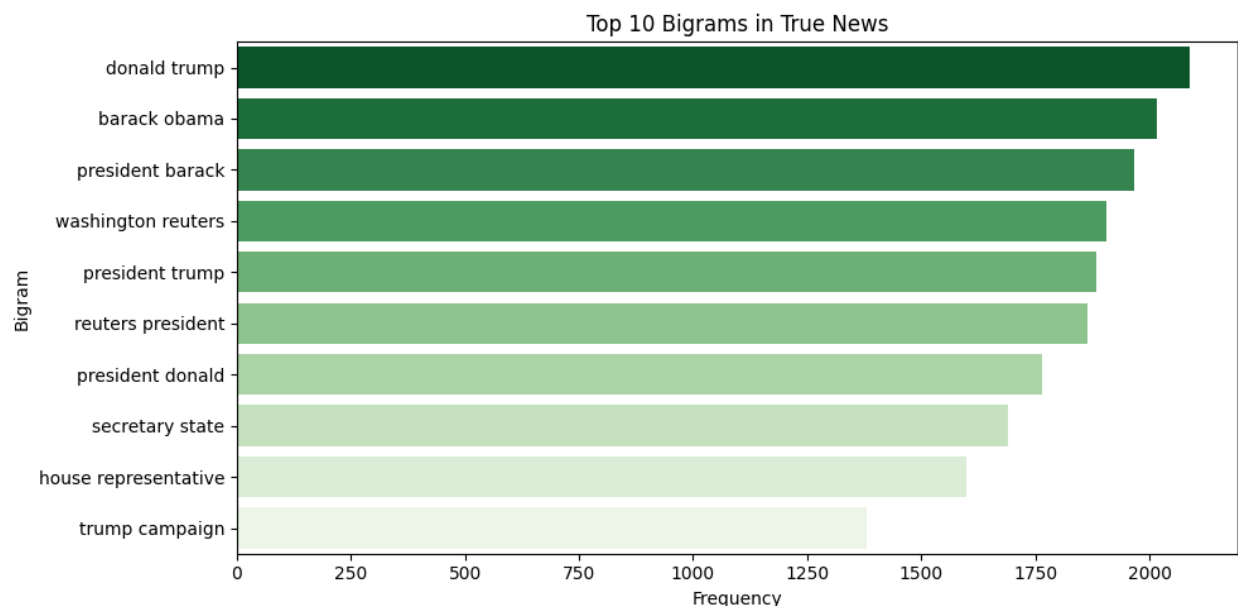


Figure 5: Top 10 Bigrams in True News (Training Set) Bar Chart

Observation:

- This horizontal bar chart presents the top 10 most frequent bigrams in true news.
- Key observations include:
 - "donald trump" is the most dominant bigram, underscoring his extensive coverage.
 - Other presidential bigrams like "barack obama," "president barack," "president trump," and "president donald" are highly frequent, highlighting a

strong focus on political leaders.

- "washington reuters" and "reuters president" suggest Reuters' prominence as a news source or subject.
- "secretary state" and "house representative" indicate frequent mentions of specific government roles.
- "trump campaign" points to coverage of political campaigns.
- In summary, true news bigrams are heavily concentrated on political figures, particularly presidents, and their associated activities, along with references to news sources and governmental roles.

Top 10 Trigrams (True News - Training Set):

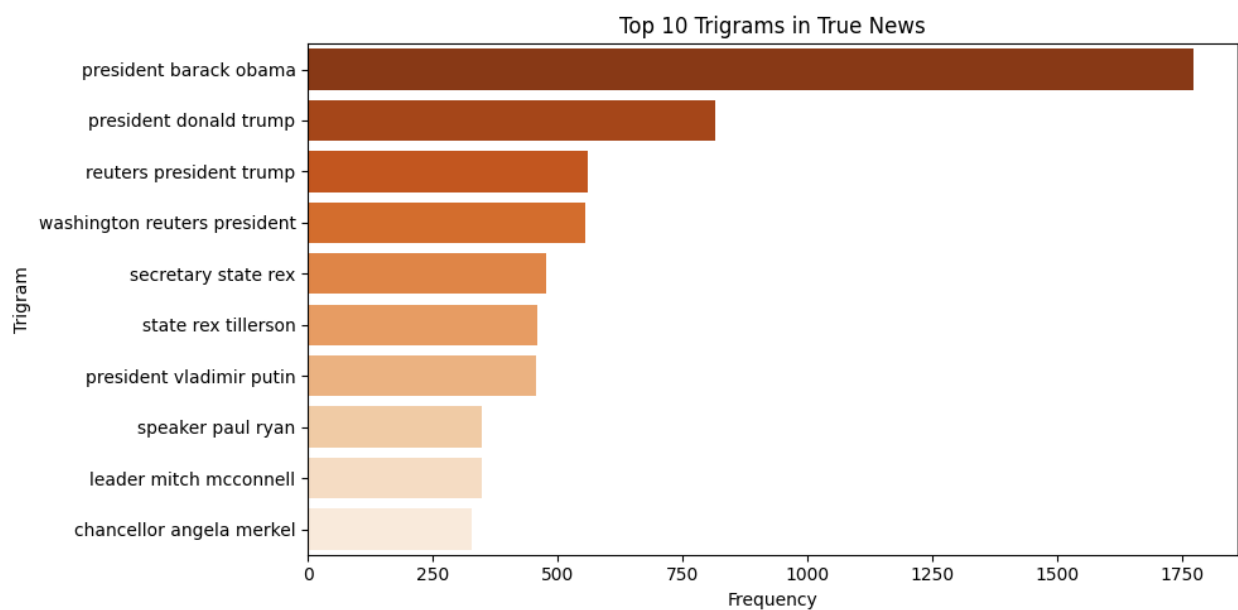


Figure 6: Top 10 Trigrams in True News (Training Set) Bar Chart

Observation:

- This horizontal bar chart presents the top 10 most frequent trigrams in true news.
- Key observations include:
 - "president barack obama" and "president donald trump" are most frequent, highlighting a strong focus on full presidential names.
 - "reuters president trump" and "washington reuters president" show Reuters' involvement in presidential news.
 - "secretary state rex" and "state rex tillerson" point to specific official mentions.
 - "president vladimir putin," "speaker paul ryan," "leader mitch mcconnell," and "chancellor angela merkel" indicate coverage of other significant political leaders.

- In summary, true news trigrams heavily feature full names and titles of prominent political figures, domestic and international, alongside Reuters' role in presidential and governmental reporting.

Top 10 Unigrams (Fake News - Training Set):

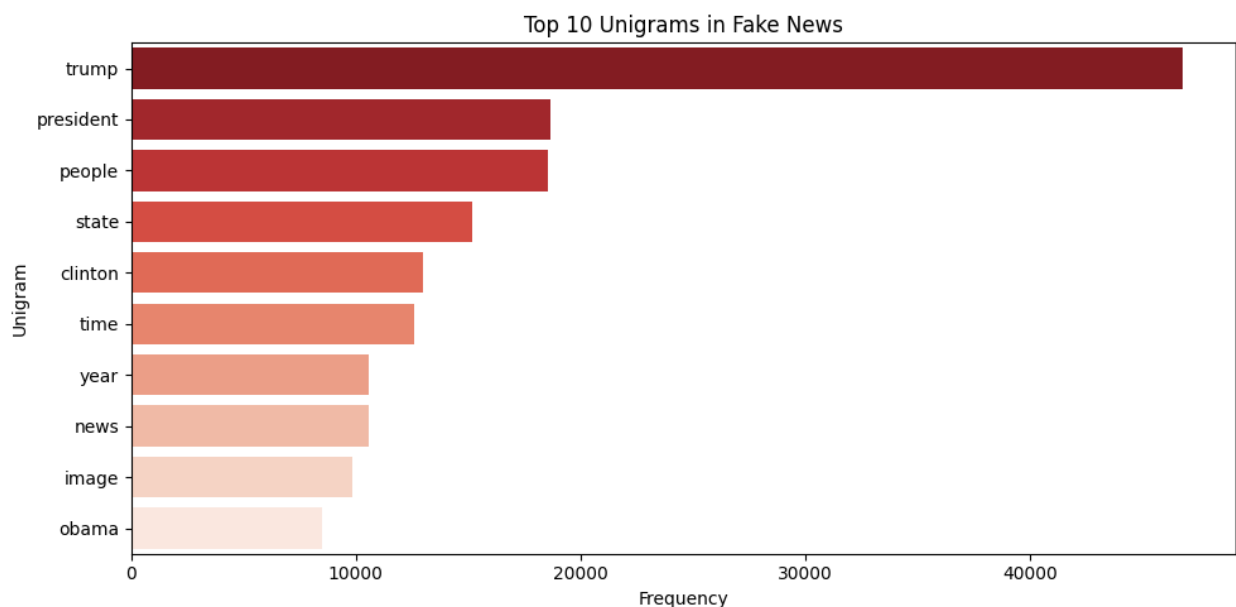


Figure 7: Top 10 Unigrams in Fake News (Training Set) Bar Chart

Observation:

- This horizontal bar chart presents the top 10 most frequent unigrams in fake news.
- Key observations include:
 - "trump" is overwhelmingly dominant, indicating a central focus.
 - "president," "people," and "state" are frequent, suggesting political and general societal discussion.
 - "clinton" and "obama" also appear, though less frequently than "trump."
 - "time," "year," and "news" are common terms, mimicking legitimate news.
 - "image" is notable, possibly suggesting visual content discussions.

- In summary, fake news unigrams heavily focus on "trump," other political figures, and general news terms, with "image" being a distinct inclusion.

Top 10 Bigrams (Fake News - Training Set):

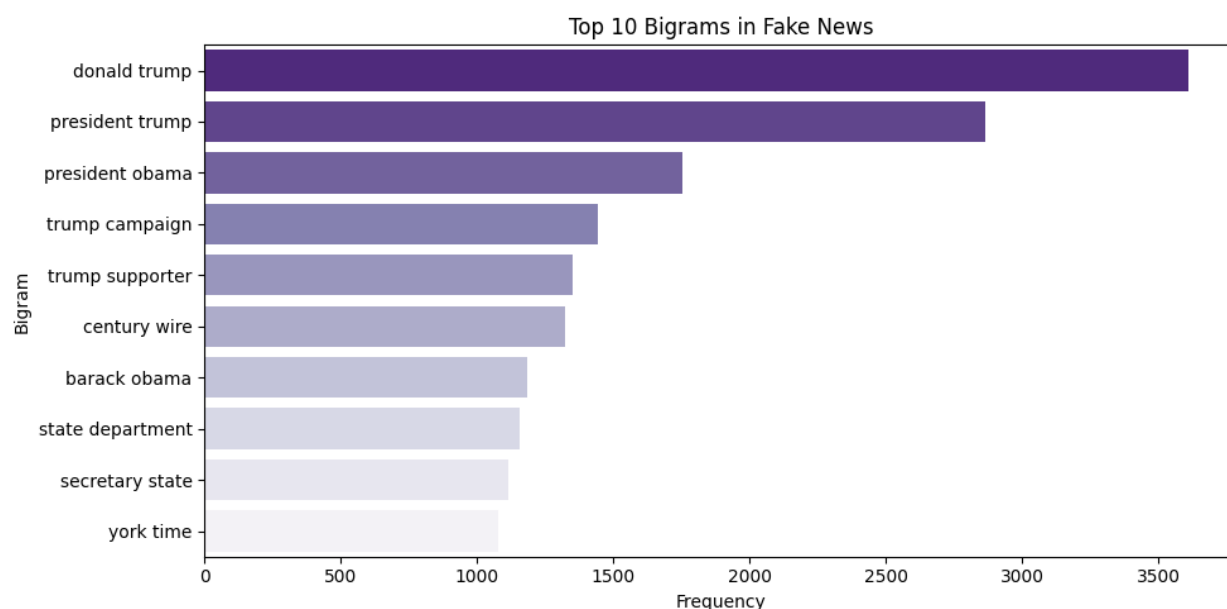


Figure 8: Top 10 Bigrams in Fake News (Training Set) Bar Chart

Observation:

- This horizontal bar chart displays the top 10 most frequent bigrams in fake news.
- Key observations include:
 - "donald trump" is the most dominant bigram, significantly surpassing others, reinforcing the strong focus on this individual.
 - "president trump" and "president obama" are also prominent, indicating a clear emphasis on presidential figures.
 - "trump campaign" and "trump supporter" highlight narratives specifically related to Trump's political activities and his base.
 - "century wire" suggests a particular source or entity frequently mentioned in fake news.
 - "barack obama" also appears, indicating continued discussion of the former president.
 - "state department" and "secretary state" point to governmental and official mentions.
 - "york time" could be a misspell or a specific reference often occurring in fake news.
- In summary, fake news bigrams are heavily centered around "donald trump" and other presidential figures, with specific mentions of Trump's campaign and

supporters, along with references to certain news sources or entities and government departments.

Top 10 Trigrams (Fake News - Training Set):

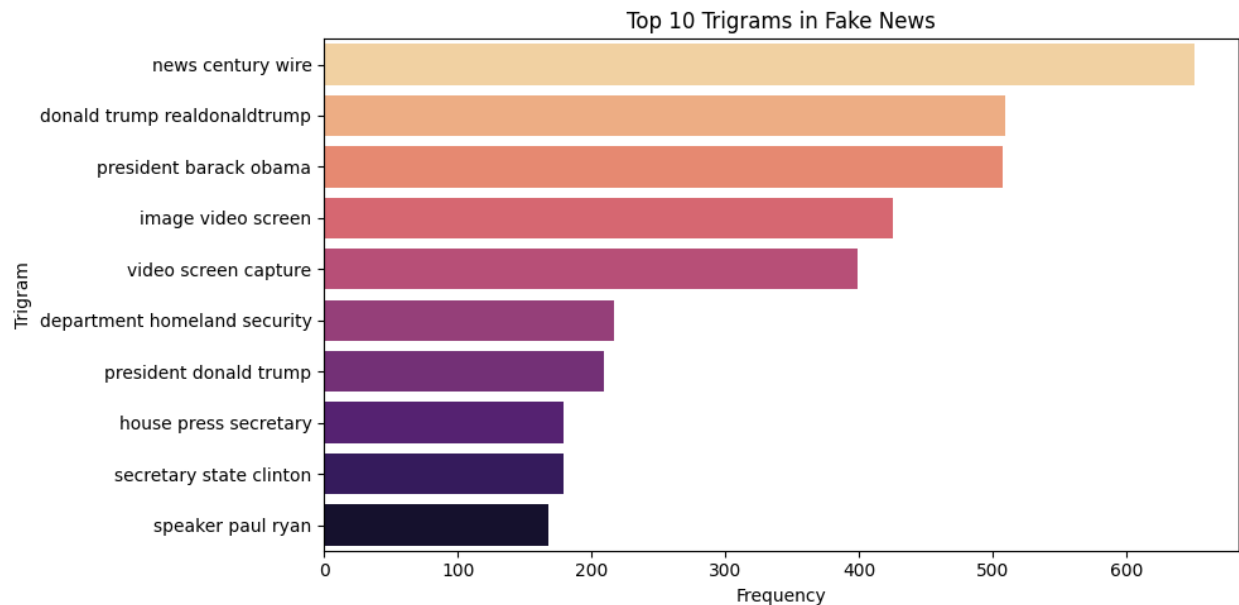


Figure 9: Top 10 Trigrams in Fake News (Training Set) Bar Chart

Observation:

- This horizontal bar chart displays the top 10 most frequent trigrams in fake news.
- Key observations include:
 - "news century wire" is the most frequent trigram, indicating a prominent source or phrase in fake news content.
 - "donald trump realdonaldtrump" suggests the direct use of social media handles or a specific emphasis on Trump's online presence in fake news.
 - "president barack obama" and "president donald trump" show a continued focus on presidential figures.
 - "image video screen" and "video screen capture" are notable, suggesting that fake news often involves or discusses visual media, possibly to fabricate or misrepresent content.
 - "department homeland security," "house press secretary," "secretary state clinton," and "speaker paul ryan" point to frequent mentions of specific government entities, roles, and officials, potentially in critical or conspiratorial contexts.
- In summary, fake news trigrams reveal a strong emphasis on specific news sources ("news century wire"), unique online identifiers ("realdonaldtrump"), prominent political figures, and a notable focus on visual media and specific

government departments and roles, suggesting a blend of political commentary and potentially manipulated content.

5.2 EDA on Validation Data

Character Length Analysis:

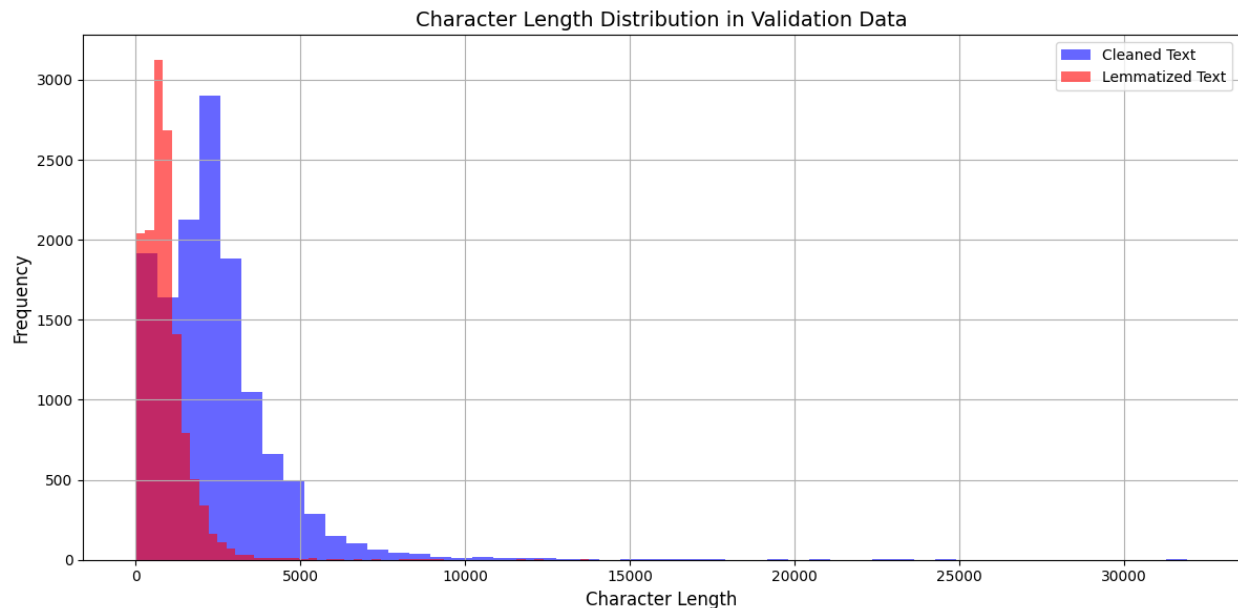


Figure 10: Histogram to visualise character lengths for Validation Data

Observation:

- This histogram shows the character length distribution for "Cleaned Text" and "Lemmatized Text" in the validation data.
- Both distributions are right-skewed.
- "Lemmatized Text" is significantly shorter, peaking around 500 characters, indicating effective length reduction by lemmatization.
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- There is overlap, but lemmatization clearly condenses texts to shorter lengths.
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Top Words by Frequency (Word Clouds)

For True News: -

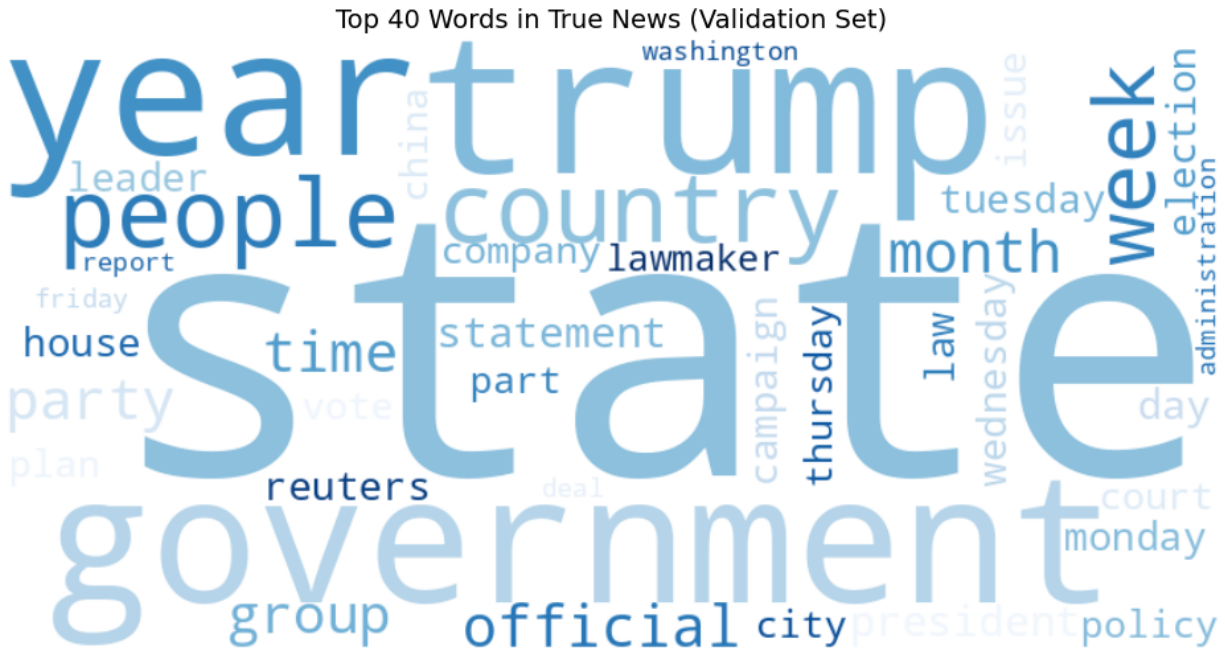


Figure 11: Top 40 Words in True News (Validation Set) Word Cloud

Observation:

- This word cloud displays the top 40 most frequent words in true news articles from the validation set.
- Key observations include:
 - "trump," "state," and "government" are the largest and most prominent words, indicating a strong focus on political figures and governmental affairs.
 - Terms like "time," "year," "month," "week," "day," and specific days of the week ("tuesday," "wednesday," "thursday," "friday," "monday") are highly visible, suggesting a strong temporal element in true news reporting.
 - Words such as "people," "country," "party," "group," "leader," "president," and "official" highlight the focus on political entities and individuals.
 - "reuters" is present, suggesting its continued relevance as a source or topic within true news.
 - Other terms like "house," "law," "vote," "election," "policy," "deal," "campaign," and "statement" reflect the common themes of political and legislative news.
- In summary, the word cloud for true news in the validation set reinforces a consistent emphasis on political figures (especially "trump"), governmental bodies, temporal aspects, and general political discourse, similar to the training set.

Top 40 Words in Fake News (Validation Set)

A word cloud visualization showing the top 40 words in a fake news validation set. The words are arranged in a circular pattern, with the most frequent words in the center and less frequent words towards the periphery. The words are in various shades of orange and red. The most prominent words are 'people', 'trump', 'president', 'time', 'day', 'country', 'state', 'medium', 'year', 'government', 'campaign', 'fact', 'man', 'thing', 'obama', 'election', 'world', 'clinton', 'american', 'party', 'part', 'today', 'video', 'life', 'woman', 'right', 'story', 'report', 'week', 'group', 'house', 'way', 'job', 'republican', 'case', 'new', 'america', 'nation', 'statement', 'donald trump', and 'government'.

way time day country statement
house group people new america nation case
report week republic election world clinton
american party part today video life woman
right story trump president state medium
year government campaign fact man thing obama

Observation:

- This word cloud displays the top 40 most frequent words in fake news articles from the validation set.
- Key observations include:
 - "trump" is overwhelmingly the largest word, demonstrating an extremely strong and consistent focus on this individual in fake news.
 - "people" and "time" are also very prominent, reflecting general conversational and temporal elements often found in sensationalized content.
 - "president," "clinton," and "obama" appear, indicating that other political figures are also central to fake news narratives, though less so than "trump."
 - Words like "country," "world," "state," and "government" suggest a global and national scope, often framed within a specific political agenda.
 - Terms such as "video," "image," "report," "fact," and "story" might be used to present or refer to potentially misleading or fabricated content.
 - "campaign," "group," "republican," "man," and "woman" indicate a focus on political movements, demographics, and gender.
- In summary, the word cloud for fake news in the validation set confirms an intense concentration on "trump," alongside other political figures, general terms, and specific words that hint at the manipulative or sensational nature of the

content, consistent with observations from the training set.

N-gram Analysis (Bar Charts)

Top 10 Unigrams (True News - Validation Set):

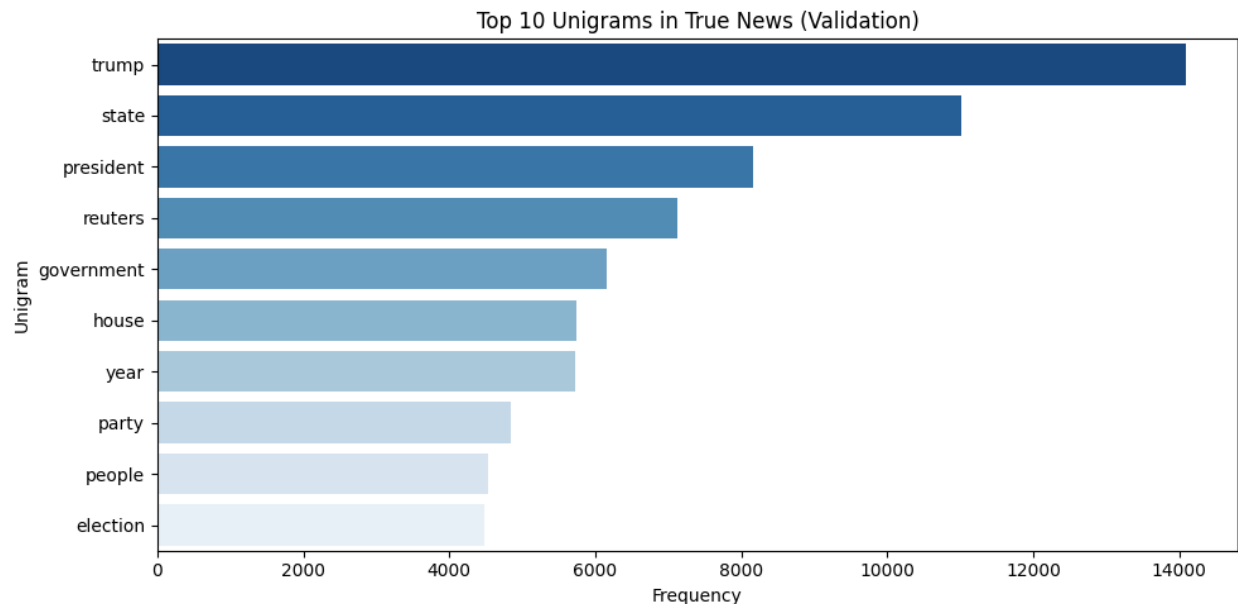


Figure 13: Top 10 Unigrams in True News (Validation Set) Bar Chart

Observation:

- This horizontal bar chart displays the top 10 unigrams by frequency in true news articles from the validation set.
- Key observations include:
 - "trump" is the most frequent unigram, consistently appearing as a central figure in true news.
 - "state," "president," and "government" are also highly frequent, reinforcing the strong emphasis on political leadership and governmental affairs.
 - "reuters" is a prominent unigram, indicating its continued significance as a news source or subject within true news.
 - Other terms like "house," "year," "party," "people," and "election" are frequently mentioned, reflecting common themes in political and societal news.
- In summary, the unigram analysis of true news in the validation set confirms a consistent focus on "trump," political institutions and figures, and the recurring presence of "reuters," aligning with observations from the training data.

Top 10 Bigrams (True News - Validation Set):

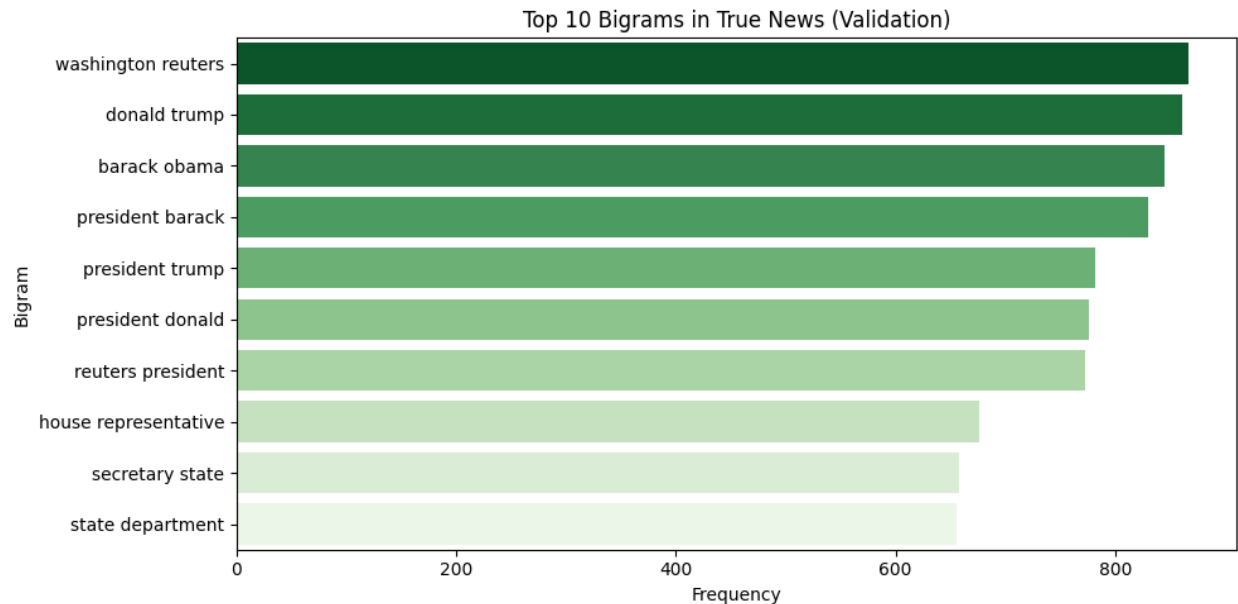


Figure 14: Top 10 Bigrams in True News (Validation Set) Bar Chart

Observation:

- This horizontal bar chart displays the top 10 most frequent bigrams in true news articles from the validation set.
- Key observations include:
 - "washington reuters" is the most frequent bigram, indicating a strong connection between Reuters and news originating from Washington D.C.
 - "donald trump," "barack obama," "president barack," "president trump," and "president donald" are highly prominent, reinforcing the focus on presidential figures by their full names or titles.
 - "reuters president" further highlights the association of Reuters with presidential reporting.
 - "house representative," "secretary state," and "state department" indicate frequent mentions of specific governmental roles and entities.
- In summary, the bigram analysis of true news in the validation set consistently shows a heavy emphasis on prominent political figures (especially presidents), the significant role of Reuters in reporting on Washington and presidential affairs, and mentions of key governmental positions.

Top 10 Trigrams (True News - Validation Set):

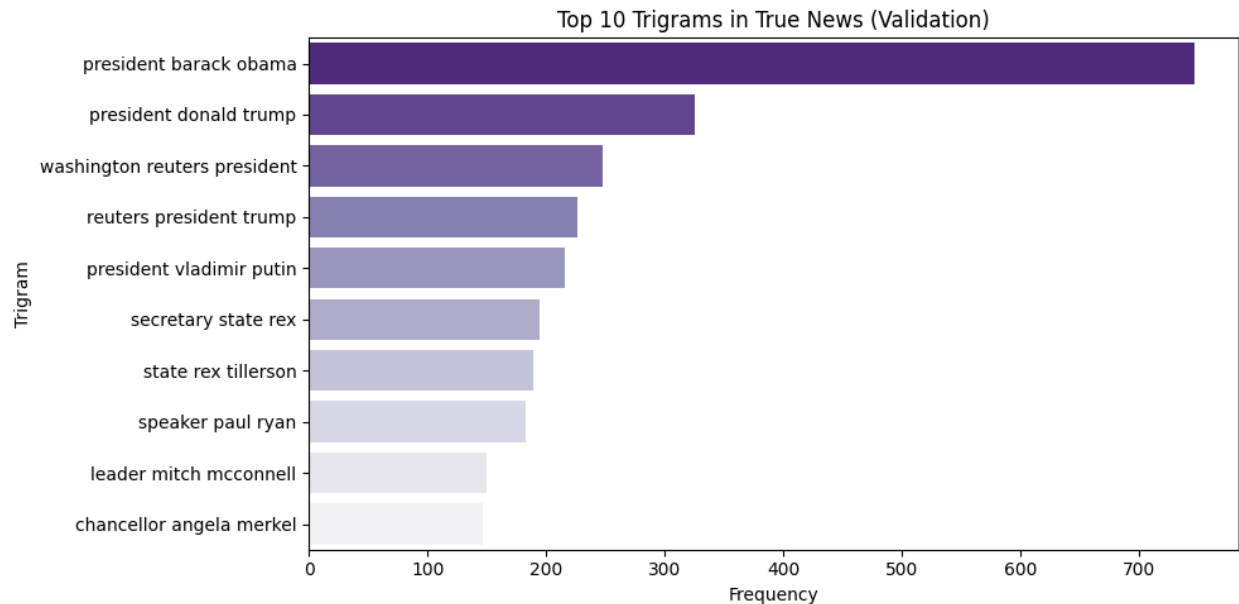


Figure 15: Top 10 Trigrams in True News (Validation Set) Bar Chart

Observation:

- This horizontal bar chart displays the top 10 most frequent trigrams in true news articles from the validation set.
- Key observations include:
 - "president barack obama" is the most frequent trigram, followed closely by "president donald trump," indicating a strong focus on full names and titles of prominent presidents.
 - "washington reuters president" and "reuters president trump" reinforce the significant connection between Reuters, Washington, and presidential news.
 - "president vladimir putin" highlights coverage of international leaders.
 - "secretary state rex" and "state rex tillerson" point to specific mentions of government officials and their titles (e.g., Rex Tillerson as Secretary of State).
 - "speaker paul ryan," "leader mitch mcconnell," and "chancellor angela merkel" indicate frequent mentions of other key political figures, both domestic and international.
- In summary, the trigram analysis of true news in the validation set consistently shows a primary focus on the full names and titles of influential political figures (both US and international), and the notable role of Reuters in reporting on presidential and governmental matters.

Top 10 Unigrams (Fake News - Validation Set):

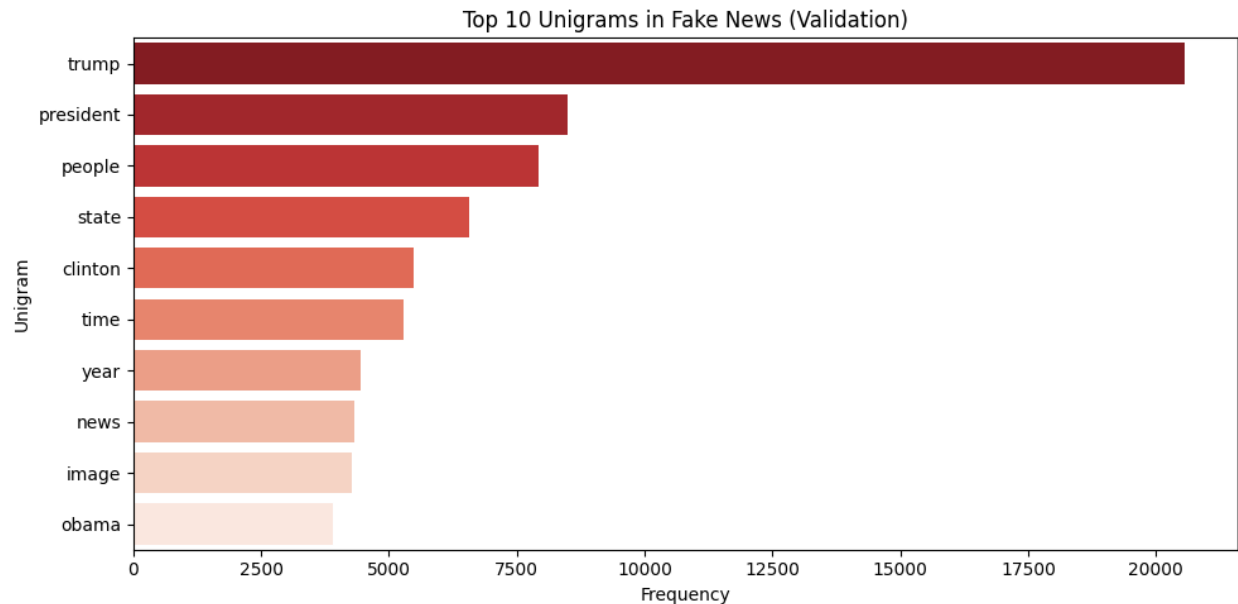


Figure 16: Top 10 Unigrams in Fake News (Validation Set) Bar Chart

Observation:

- This horizontal bar chart presents the top 10 most frequent unigrams in fake news.
- Key observations include:
 - "trump" is overwhelmingly dominant, indicating a central focus.
 - "president," "people," and "state" are frequent, suggesting political and general societal discussion.
 - "clinton" and "obama" also appear, though less frequently than "trump."
 - "time," "year," and "news" are common terms, mimicking legitimate news.
 - "image" is notable, possibly suggesting visual content discussions.
- In summary, fake news unigrams heavily focus on "trump," other political figures, and general news terms, with "image" being a distinct inclusion.

Top 10 Bigrams (Fake News - Validation Set):

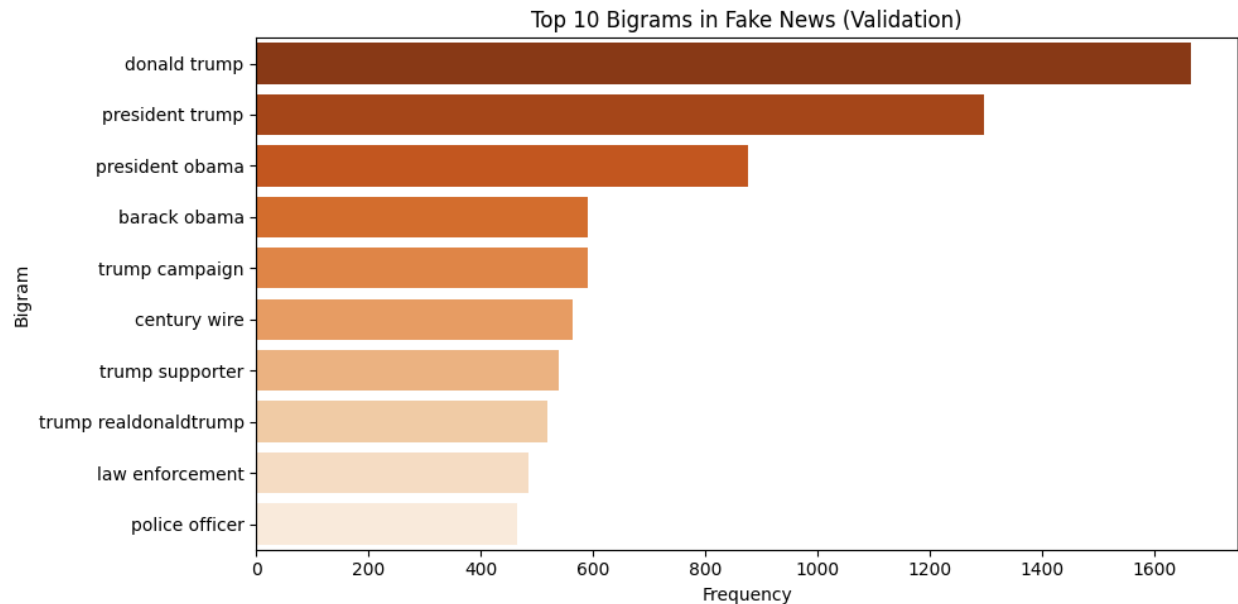


Figure 17: Top 10 Bigrams in Fake News (Validation Set) Bar Chart

Observation:

- This horizontal bar chart displays the top 10 most frequent bigrams in fake news.
- Key observations include:
 - "donald trump" is the most dominant bigram, significantly surpassing others, reinforcing the strong focus on this individual.
 - "president trump" and "president obama" are also prominent, indicating a clear emphasis on presidential figures.
 - "barack obama" also appears, continuing the discussion of former presidents.
 - "trump campaign," "century wire," and "trump supporter" highlight specific narratives related to Trump's political activities, a particular source, and his base.
 - "trump realdonaldtrump" is notable, suggesting the use of social media handles or a focus on Trump's online presence.
 - "law enforcement" and "police officer" indicate frequent mentions related to legal and law enforcement themes, which could be part of specific fake news narratives.
- In summary, fake news bigrams in the validation set are highly concentrated around "donald trump" and other presidential figures, with strong emphasis on Trump's political activities and online presence, alongside mentions of specific sources and law enforcement topics.

Top 10 Trigrams (Fake News - Validation Set):

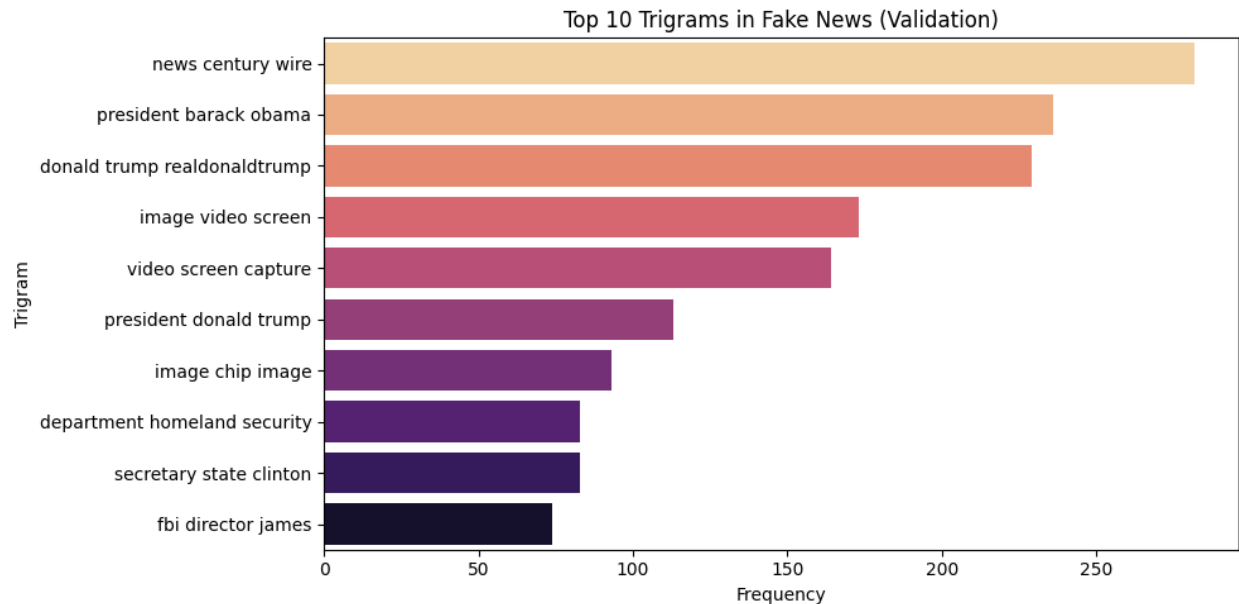


Figure 18: Top 10 Trigrams in Fake News (Validation Set) Bar Chart

Observation:

- This horizontal bar chart displays the top 10 most frequent trigrams in fake news.
- Key observations include:
 - "news century wire" is the most frequent trigram, indicating a prominent source or phrase consistently appearing in fake news.
 - "president barack obama" and "donald trump realdonaldtrump" are highly prominent, highlighting the focus on full names of political figures and the specific mention of Trump's social media handle.
 - "image video screen" and "video screen capture" are notable, suggesting that discussions about visual media, possibly manipulated, are common in fake news.
 - "president donald trump" is also present, further emphasizing his centrality.
 - "image chip image" is a peculiar trigram, potentially related to discussions about digital imagery or technology in fake news.
 - "department homeland security," "secretary state clinton," and "fbi director james" indicate frequent mentions of specific government entities, roles, and officials (e.g., James Comey as FBI Director).
- In summary, fake news trigrams in the validation set show a strong emphasis on specific news sources, political figures (with a unique focus on Trump's online presence), and a notable recurring theme around visual media, alongside mentions of key government departments and officials.

6. Feature Extraction

For feature extraction, we used Word2Vec to convert the processed text into numerical vectors. This method helps capture the relationships between words based on their context — so words with similar meanings end up close to each other in the vector space.

To represent each news article, we averaged the Word2Vec vectors of all the words in that article. This gave us a fixed-length vector for each one, capturing its overall meaning. These semantic features were then used as input for our machine learning models, allowing them to learn from the content of the text, not just its structure or keywords.

7. Model Training and Evaluation

Three supervised learning models were trained and evaluated on the validation dataset:

- 7.1. Logistic Regression,
- 7.2. Decision Tree, and
- 7.3. Random Forest.

7.1 Logistic Regression Performance on Validation Data

Accuracy: 0.9268

Precision: 0.9208

Recall: 0.9262

F1-Score: 0.9235

Classification Report for Logistic Regression model:

Class	Precision	Recall	F1-Score	Support
0	0.93	0.93	0.93	7048
1	0.92	0.93	0.92	6425
Accuracy			0.93	13473
Macro Avg.	0.93	0.93	0.93	13473
Weighted Avg.	0.93	0.93	0.93	13473

Observation:

- The Logistic Regression model performs strongly on the validation data.

- Key observations include:
 - **Accuracy:** High at 0.9268, indicating nearly 93% correct predictions.
 - **Precision:** 0.9208 overall, with 0.93 for class 0 and 0.92 for class 1, showing high confidence in positive predictions.
 - **Recall:** 0.9262 overall, with 0.93 for both classes, indicating effective capture of actual instances.
 - **F1-Score:** 0.9235 overall, with 0.93 for class 0 and 0.92 for class 1, confirming a good balance between precision and recall.
 - **Support:** Balanced class distribution in the validation set (7048 for class 0, 6425 for class 1).
- In summary, the Logistic Regression model is robust and reliable for distinguishing true and fake news, showing consistently high performance across all key metrics for both classes.

7.2 Decision Tree Performance on Validation Data

Accuracy: 0.8484

Precision: 0.8508

Recall: 0.8272

F1-Score: 0.8389

Classification Report for Decision Tree model:

Class	Precision	Recall	F1-Score	Support
0	0.85	0.87	0.86	7048
1	0.85	0.83	0.84	6425
Accuracy			0.85	13473
Macro Avg.	0.85	0.85	0.85	13473
Weighted Avg.	0.85	0.85	0.85	13473

Observation:

- The Decision Tree model exhibits moderate performance on the validation data.
- Key observations include:
 - **Accuracy:** At 0.8484, it correctly predicts about 84.84% of the samples.
 - **Precision:** Overall 0.8508, with similar values for both classes (0.85 for class 0, 0.85 for class 1), indicating a fair rate of correct positive predictions.
 - **Recall:** Overall 0.8272, with 0.87 for class 0 and 0.83 for class 1, suggesting it

is slightly better at identifying class 0 instances.

- **F1-Score:** Overall 0.8389, with 0.86 for class 0 and 0.84 for class 1, indicating a reasonable balance between precision and recall, but slightly lower than Logistic Regression.
- **Support:** A balanced distribution of samples across classes (7048 for class 0, 6425 for class 1).
- In summary, the Decision Tree model provides decent but not outstanding performance, with slightly lower metrics across the board compared to the Logistic Regression model, particularly in recall for class 1.

7.3 Random Forest Performance on Validation Data

Accuracy: 0.9236

Precision: 0.9306

Recall: 0.9074

F1-Score: 0.9188

Classification Report for Random Forest model:

Class	Precision	Recall	F1-Score	Support
0	0.92	0.94	0.93	7048
1	0.93	0.91	0.92	6425
Accuracy			0.92	13473
Macro Avg.	0.92	0.92	0.92	13473
Weighted Avg.	0.92	0.92	0.92	13473

Observation:

- The Random Forest model shows strong performance on the validation data.
- Key observations include:
 - **Accuracy:** High at 0.9236, indicating correct predictions for over 92% of samples.
 - **Precision:** Overall 0.9306, with 0.92 for class 0 and 0.93 for class 1, showing strong confidence in positive predictions for both.
 - **Recall:** Overall 0.9074, with 0.94 for class 0 and 0.91 for class 1, suggesting it's slightly better at identifying class 0.
 - **F1-Score:** Overall 0.9188, with 0.93 for class 0 and 0.92 for class 1, indicating a good balance between precision and recall, especially strong for class 0.

- **Support:** Balanced class distribution (7048 for class 0, 6425 for class 1).
- In summary, the Random Forest model is a strong performer, comparable to Logistic Regression, with high accuracy and well-balanced precision and recall across both classes, though slightly favoring recall for class 0.

8. Outcomes and Conclusion

We built a semantic classification model for detecting fake news using Word2Vec embeddings. The process began with data cleaning and thorough text preprocessing, where we applied lemmatization and POS tagging to focus on informative components like nouns. This step was essential in turning the raw text into structured input suitable for semantic analysis. Alongside this, exploratory data analysis helped us understand how the language used in fake news differs from that in real news articles.

For the modelling phase, we compared Logistic Regression, Random Forest, and Decision Tree classifiers. Logistic Regression gave the best performance, achieving an accuracy of 92.68% and an F1-score of 92.35%. Random Forest followed closely with 92.36% accuracy and an F1-score of 91.88%. The Decision Tree model lagged behind in both metrics.

We prioritized the F1-score in evaluating our models because it balances precision and recall. In fake news detection, minimizing both false positives and false negatives is important, as errors can lead to serious misinformation issues. Interestingly, Logistic Regression—despite its simplicity—outperformed more complex models when paired with semantically rich features generated by Word2Vec.

Overall, the approach proved effective in distinguishing fake from real news. The combination of linguistic preprocessing, semantic feature extraction, and focused model evaluation resulted in a well-rounded and interpretable solution.